

HURON LIGHT, OH, USA

WMO: 998001

Lat: 41.4005N

Lon: 82.5452W

Elev: 604

StdP: 14.38

Time Zone: -5.00 (NAE)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 2.1	(c) 8.1	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) 31.8	(k) 24.3	(l) 28.2	(m) 23.8	(n) 10.6	(o) 230	(p) 0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.3	(c) 88.5	(d) N/A	(e) 85.7	(f) N/A	(g) 82.9	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) 10.8	(p) 240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) N/A	(b) N/A	(c) N/A	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 27.9	(b) 24.5	(c) 20.7	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(4) 27.9	(5) 24.5	(6) 20.7	(7) WB	(8) N/A	(9) N/A	(10) N/A	(11) N/A	(12) N/A	(13) N/A	(14) N/A	(15) N/A	(16) N/A	(17) N/A	(18) N/A	(19) N/A	(20) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.6	26.0	28.9	36.9	49.7	61.1	70.1	74.0	71.8	66.5	54.8	43.0	34.6		
	DBStd	18.68	11.81	10.92	9.84	8.85	8.36	6.02	5.34	4.27	6.74	7.59	9.23	9.49		
	HDD50	2660	748	596	427	113	6	0	0	0	0	35	250	485		
	HDD65	5756	1209	1012	870	469	180	19	1	3	59	330	662	942		
	CDD50	3232	4	4	22	103	350	602	745	677	496	183	38	8		
	CDD65	853	0	0	0	9	60	171	281	215	104	13	0	0		
	CDH74	5167	0	0	1	93	384	979	2025	1056	579	50	0	0		
	CDH80	1283	0	0	0	14	98	266	542	190	167	6	0	0		
(14) Wind	WSAvg	10.4	11.9	10.9	11.3	11.7	9.8	9.3	8.5	8.6	9.5	10.8	11.1	11.0		
(15) Precipitation	PrecAvg	36.1	2.3	2.1	2.4	3.7	3.5	4.1	3.7	3.2	3.3	2.6	2.7	2.6		
	PrecMax	44.4	5.2	4.9	4.7	5.6	6.4	7.7	8.0	7.8	7.7	3.9	5.6	6.6		
	PrecMin	25.1	0.6	0.5	1.0	1.8	0.8	1.6	1.3	0.4	1.0	0.7	0.7	0.9		
	PrecStd	5.5	1.1	1.1	1.0	1.0	1.6	1.6	1.7	1.9	1.7	0.8	1.0	1.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.8	65.4	70.2	82.4	87.4	90.6	93.0	88.9	90.2	78.4	68.8	63.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	DB	55.7	56.0	63.8	76.4	82.5	86.5	88.7	84.9	85.3	74.7	64.2	58.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	DB	48.5	49.5	59.4	71.2	78.3	83.0	85.8	81.9	79.9	70.3	60.9	54.0		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	DB	41.8	43.1	53.2	65.6	74.3	79.9	82.8	79.5	76.7	66.4	57.4	48.4		
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.4	11.9	14.2	13.1	11.8	12.3	11.7	12.2	13.0	11.9	10.7		
		MCDBR	18.8	20.6	20.8	24.6	20.3	16.6	17.0	15.9	17.6	18.5	17.3	17.4		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(40) Clear-Sky Solar Irradiance	taub		0.308	0.335	0.359	0.399	0.433	0.449	0.446	0.433	0.387	0.367	0.329	0.315		
	taud		2.401	2.298	2.310	2.256	2.203	2.206	2.236	2.258	2.361	2.396	2.464	2.435		
	Ebn at Noon		267	274	281	277	269	264	263	263	268	260	257	254		
	Edh at Noon		26	33	37	42	45	45	43	41	35	30	24	23		
(44) All-Sky Solar Radiation	RadAvg		447	705	1078	1453	1721	1910	1924	1695	1364	889	561	376		
	RadStd		45	98	95	112	143	129	109	66	119	69	83	42		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+182	N/A

Nomenclature: See separate page